CCA tracking metrics for climate–vegetation comparisons

Working calculation guide: Anthropocene paired resurveys and Holocene 500-year time series

Core notation. For any comparison between t_1 and t_2, define the climate-predicted CCA position $\boldsymbol{P}_t = (P_{1,t}, P_{2,t})$ from LC scores and the observed vegetation position $\boldsymbol{O}_t = (O_{1,t}, O_{2,t})$ from WA scores. Then

$$\Delta\boldsymbol{P} = \boldsymbol{P}_{t_2} - \boldsymbol{P}_{t_1}, \qquad \Delta\boldsymbol{O} = \boldsymbol{O}_{t_2} - \boldsymbol{O}_{t_1}.$$

All metrics below should be calculated from these two displacement vectors **before averaging or summarizing across plots, age draws, or bin offsets**.

1. Basic displacement and rate metrics

Metric	Calculation	Interpretation
Predicted displacement	$D_P = \|\Delta\boldsymbol{P}\|$	How far climate predicts the community should move in CCA space.
Observed turnover	$D_O = \|\Delta\boldsymbol{O}\|$	How far vegetation actually moved in CCA space.
Predicted rate	$R_P = D_P/\Delta t_P \times 100$	Climate-predicted compositional movement in CCA units per century.
Observed turnover rate	$R_O = D_O/\Delta t_O \times 100$	Observed vegetation turnover in CCA units per century.

Important: use the appropriate elapsed time for each data stream. In the Anthropocene, survey and climate-window intervals may differ. In the Holocene 500-year analysis, $\Delta t_O = \Delta t_P = 500$ yr.

2. Magnitude and rate tracking

Magnitude tracking ratio

$$M = \frac{D_O}{D_P}.$$

$M = 1$ means equal displacement magnitude; $M < 1$ means vegetation moved less; $M > 1$ means vegetation moved farther. This metric ignores direction.

Rate tracking ratio

$$M_R = \frac{R_O}{R_P}.$$

$M_R = 1$ means vegetation reorganized at the same rate magnitude as predicted by climate. This is the most direct metric for the question "did vegetation turnover keep pace with climate-predicted turnover?"

For Holocene 500-year comparisons,

$$M_R = \frac{D_O/500}{D_P/500} = \frac{D_O}{D_P} = M,$$

so magnitude tracking and rate tracking are numerically identical. In the Anthropocene they need not be identical if the time denominators differ.

Directional tracking metrics

3. Angle and directional agreement

Angle between observed and predicted movement

$$\theta = \cos^{-1}\left(\frac{\Delta \boldsymbol{O} \cdot \Delta \boldsymbol{P}}{\|\Delta \boldsymbol{O}\| \, \|\Delta \boldsymbol{P}\|}\right).$$

Interpretation: 0° = same direction; 90° = perpendicular; 180° = opposite direction.

Equivalent cosine similarity:

$$C = \cos(\theta) = \frac{\Delta \boldsymbol{O} \cdot \Delta \boldsymbol{P}}{D_O D_P}.$$

$C = 1$ = same direction; $C = 0$ = orthogonal; $C = -1$ = opposite direction.

Do not interpret the angle when either D_P or D_O is extremely small. In that case direction is unstable and a small amount of noise can generate any angle.

4. Climate-aligned component of vegetation change

Projection of observed vegetation movement onto the climate-predicted direction:

$$D_{\parallel} = \frac{\Delta \boldsymbol{O} \cdot \Delta \boldsymbol{P}}{\|\Delta \boldsymbol{P}\|}.$$

Normalized directional tracking:

$$T_{\parallel} = \frac{D_{\parallel}}{D_P} = \frac{\Delta \boldsymbol{O} \cdot \Delta \boldsymbol{P}}{\|\Delta \boldsymbol{P}\|^2} = M \cos\theta.$$

Interpretation:

- $T_{\parallel} = 1$: exactly the predicted amount of movement in the predicted direction.
- $0 < T_{\parallel} < 1$: partial progress along the climate-predicted trajectory.
- $T_{\parallel} = 0$: no net movement along the climate-predicted direction.
- $T_{\parallel} < 0$: vegetation moved opposite to climate prediction.
- $T_{\parallel} > 1$: vegetation moved farther than predicted along the climate trajectory.

Perpendicular component

$$D_{\perp} = \sqrt{D_O^2 - D_{\parallel}^2}, \qquad T_{\perp} = \frac{D_{\perp}}{D_P}.$$

This captures turnover that is not aligned with the climate-predicted trajectory. It may be useful for identifying reassembly driven by disturbance or other non-climatic processes.

Tracking error and lag metrics

5. Overall trajectory mismatch

Tracking error / trajectory mismatch

$$E = \|\Delta \boldsymbol{O} - \Delta \boldsymbol{P}\|.$$

Perfect tracking gives $E = 0$. This combines magnitude and direction, but does not identify which of them caused the mismatch.

Normalized form:

$$E_R = \frac{\|\Delta \boldsymbol{O} - \Delta \boldsymbol{P}\|}{\|\Delta \boldsymbol{P}\|}.$$

6. Climate–vegetation lag at each time and change in lag

Lag magnitude at time t:

$$L_t = \|\boldsymbol{P}_t - \boldsymbol{O}_t\|.$$

This is the instantaneous mismatch between the climate-predicted and observed community positions.

Change in lag across the interval:

$$\Delta L = L_{t_2} - L_{t_1}.$$

Interpretation: $\Delta L > 0$ = mismatch increased; $\Delta L < 0$ = mismatch decreased; $\Delta L = 0$ = mismatch magnitude stayed constant.

Note: ΔL and E are not the same quantity. ΔL compares the lengths of the two offset vectors, whereas E measures how much the offset vector itself changed.

Recommended core outputs

For the main ecological interpretation, retain these four as the primary tracking descriptors:

1. R_O and R_P: observed versus climate-predicted turnover rates;
2. M_R: rate tracking ratio;
3. θ or C: directional agreement;
4. $T_{\parallel}$: realized movement along the climate-predicted trajectory;
5. ΔL: whether climate–vegetation mismatch is opening or closing.

$D_{\perp}$ and E are useful secondary diagnostics, especially when testing whether disturbance creates turnover that is large but not climate-directed.

How to calculate these metrics in each era

Anthropocene: paired resurveys

The atomic unit is the **paired plot comparison**. For each plot i:

1. obtain $\boldsymbol{O}_{i,t_1}$ and $\boldsymbol{O}_{i,t_2}$ from WA scores;
2. obtain $\boldsymbol{P}_{i,t_1}$ and $\boldsymbol{P}_{i,t_2}$ from LC scores;
3. calculate $\Delta\boldsymbol{O}_i$ and $\Delta\boldsymbol{P}_i$;
4. calculate all distances, rates, ratios, angles, projections, tracking error, and ΔL at the plot level;
5. only then summarize across plots within MORA or NOCA (for example median, IQR, bootstrap CI if desired).

Do not calculate the site-centroid movement first and use that as the plot-level turnover estimate: the distance between centroids can be much smaller than the median of individual plot displacement magnitudes when plots move in different directions.

Holocene: current full time-series analysis only

Retain the current standardized structure: **500-year bins with 100-year bin offsets and random age draws**. The atomic calculation level is:

$$\boxed{\text{lake/site} \times \text{offset} \times \text{age draw} \times \text{consecutive 500-year interval.}}$$

For each realization r and consecutive bins $t_1 \to t_2$:

1. aggregate the macrofossil community within each 500-year bin and obtain WA positions $\boldsymbol{O}_{t_1,r}$ and $\boldsymbol{O}_{t_2,r}$;
2. obtain climate-predicted LC positions $\boldsymbol{P}_{t_1,r}$ and $\boldsymbol{P}_{t_2,r}$ for the same bins;
3. calculate $\Delta\boldsymbol{O}_r$ and $\Delta\boldsymbol{P}_r$;
4. calculate all nonlinear tracking metrics **at this realization level**: D_O, D_P, R_O, R_P, M_R, θ, $T_{\parallel}$, $D_{\perp}$, E, L_{t_1}, L_{t_2}, and ΔL;
5. summarize across age draws **within each offset**;
6. then summarize across offsets to represent sensitivity to bin phase.

Do not first average CCA positions, displacement vectors, rates, or ratios across age draws/offsets and then calculate the tracking metrics. Norms, ratios, angles, and projections are nonlinear, so the order of operations changes the result.

For every Holocene 500-year comparison,

$$R_O = D_O/500 \times 100, \qquad R_P = D_P/500 \times 100,$$

therefore $M_R = M$. Keep the rate formulation in the output because it remains directly comparable with the Anthropocene rate metric.

Practical flags

- Flag intervals with very small D_P before interpreting ratios, $T_{\parallel}$, or normalized error; division by a near-zero climate displacement is unstable.
- Flag intervals with very small D_O before interpreting angle; direction is undefined when vegetation barely moves.
- Preserve the full Holocene time series in the appendix even if later analyses group intervals into ecologically meaningful periods.